

**TRAIN
PROBLEMS?**

Basic To Advance

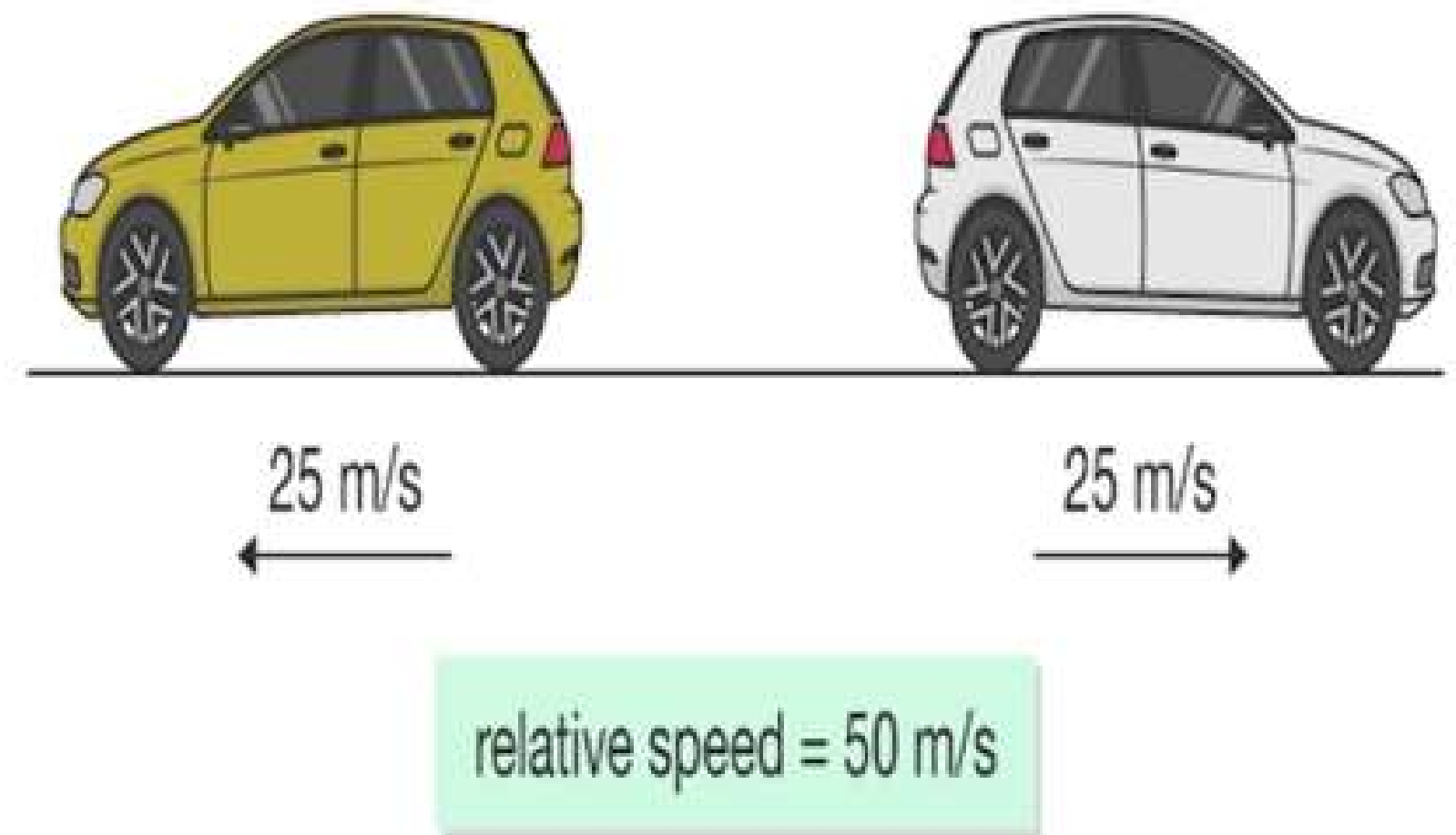
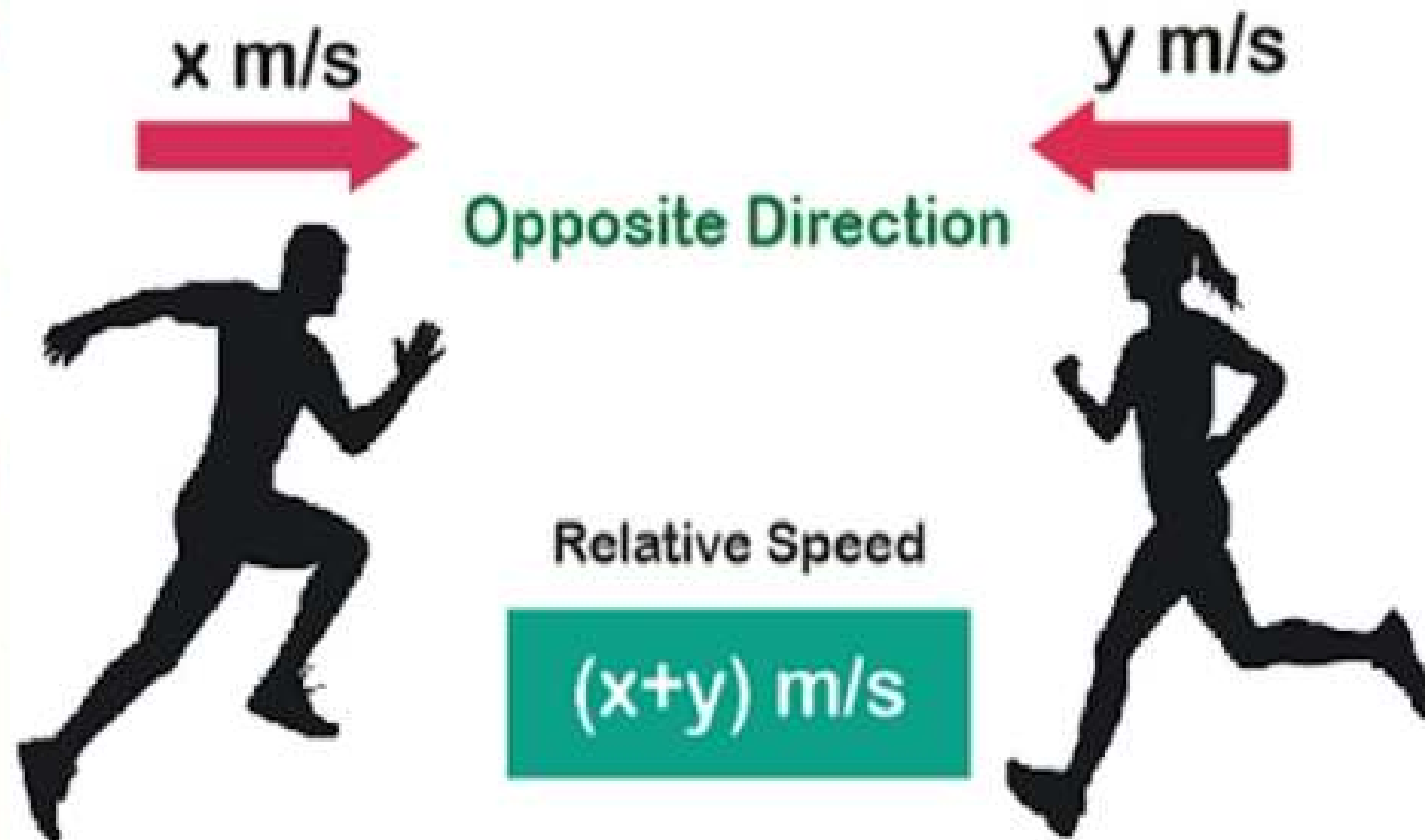
- by YASH JAIN



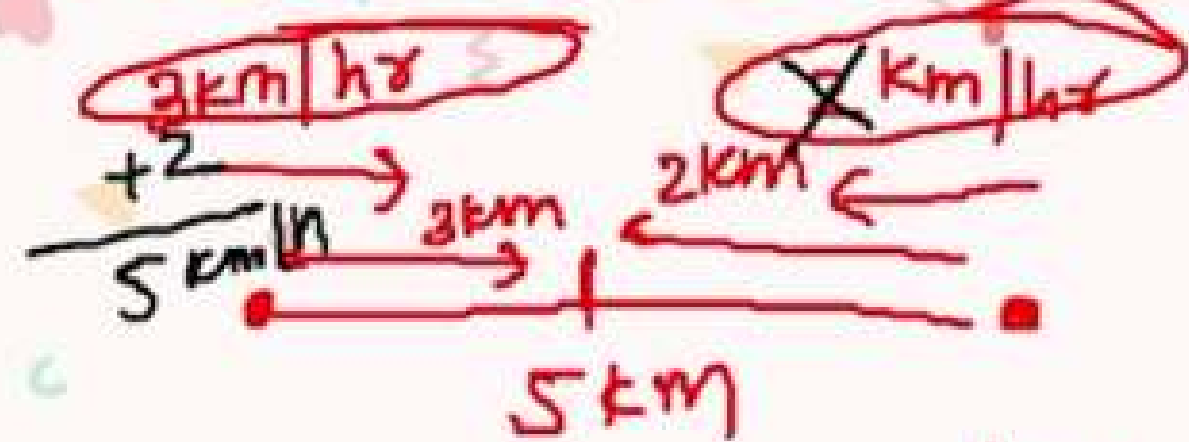
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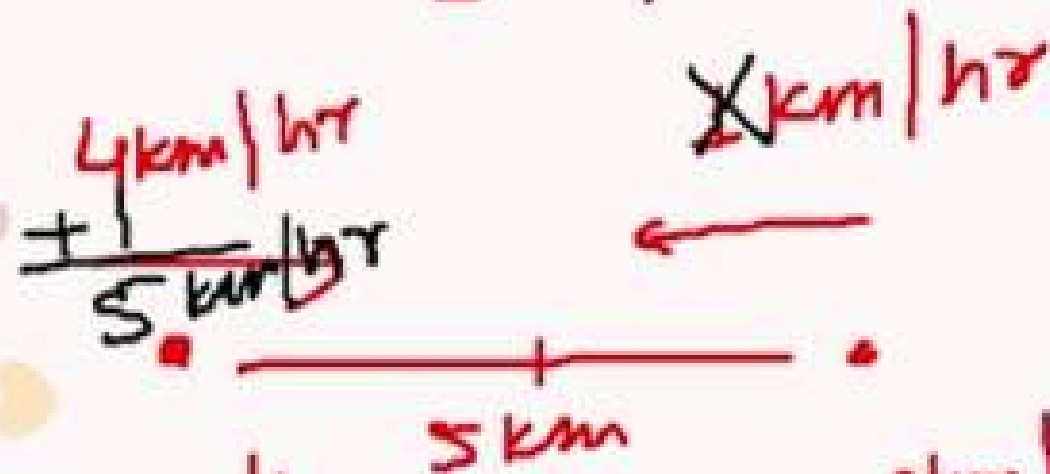
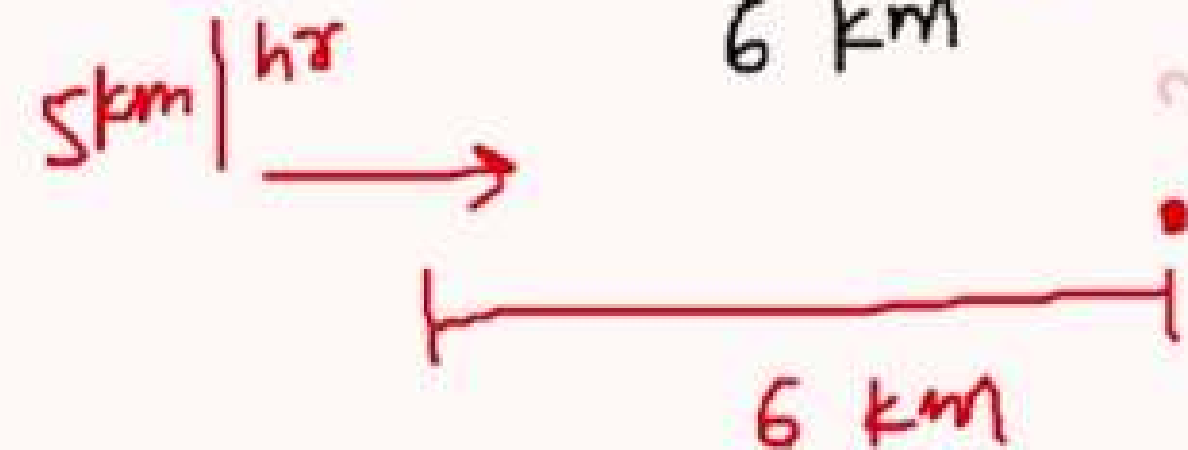
Relative Speed



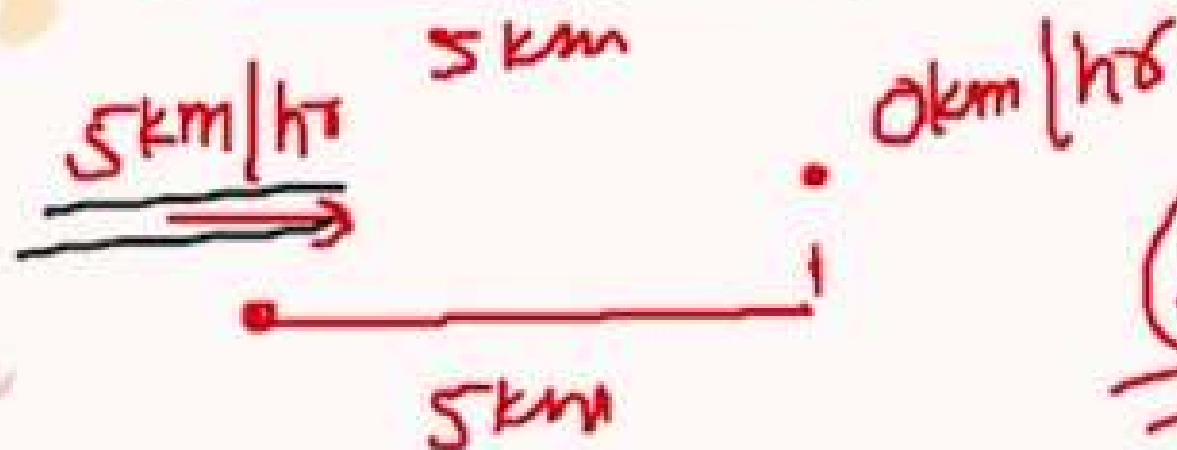
Relative Speed (Before Relationship)



1 hr



1 hr ✓



1 hr ✓

$$s = \frac{d}{t}$$

$$5 = \frac{6}{t}$$

$$t = \frac{6}{5} = 1.2 \text{ hr}$$

Relative Speed (Before Relationship)

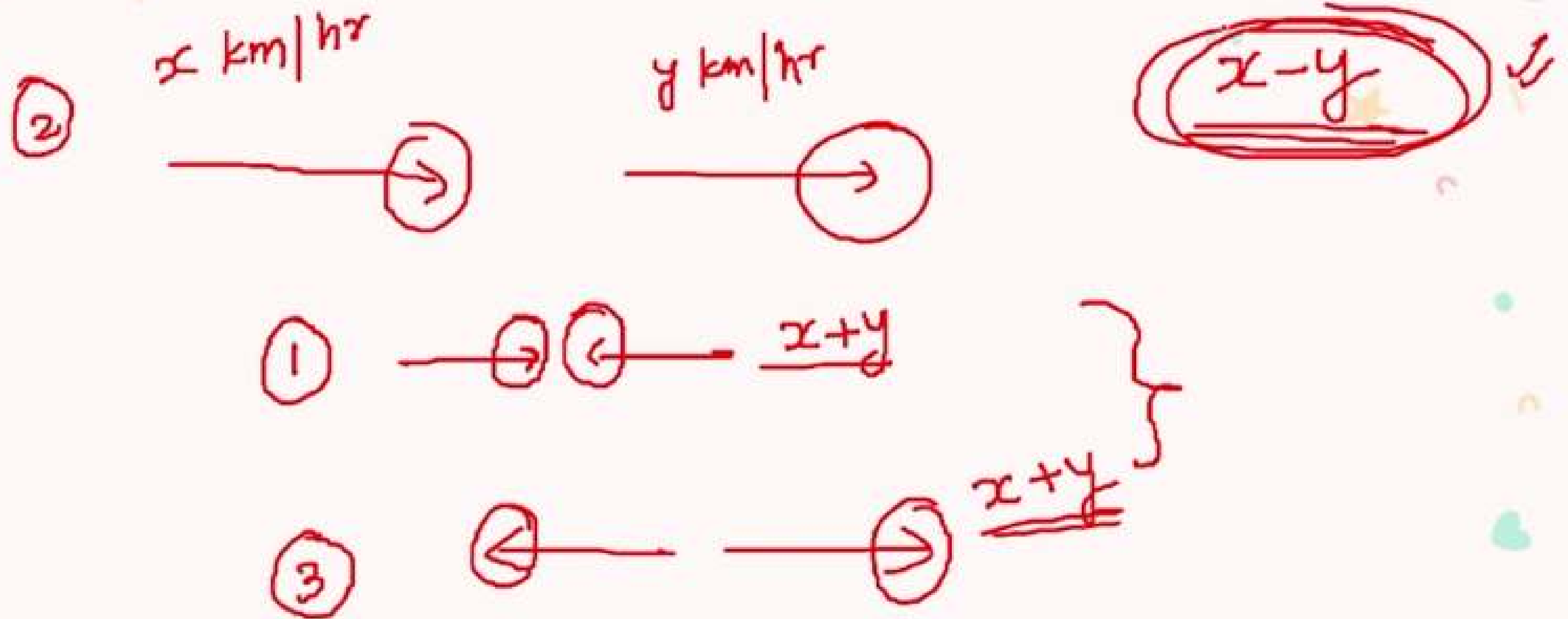


$$\boxed{\text{relative} = x + y}$$

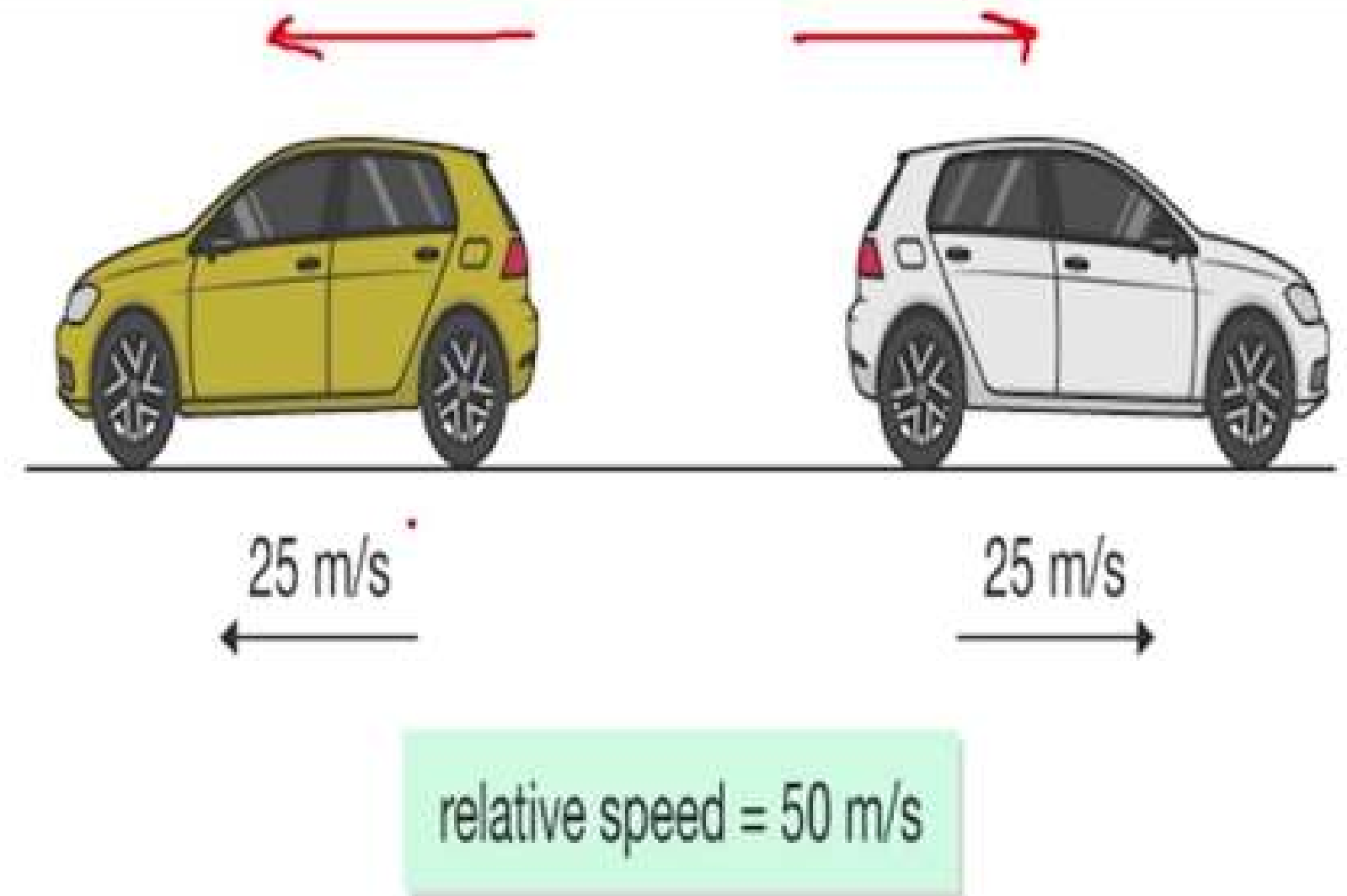
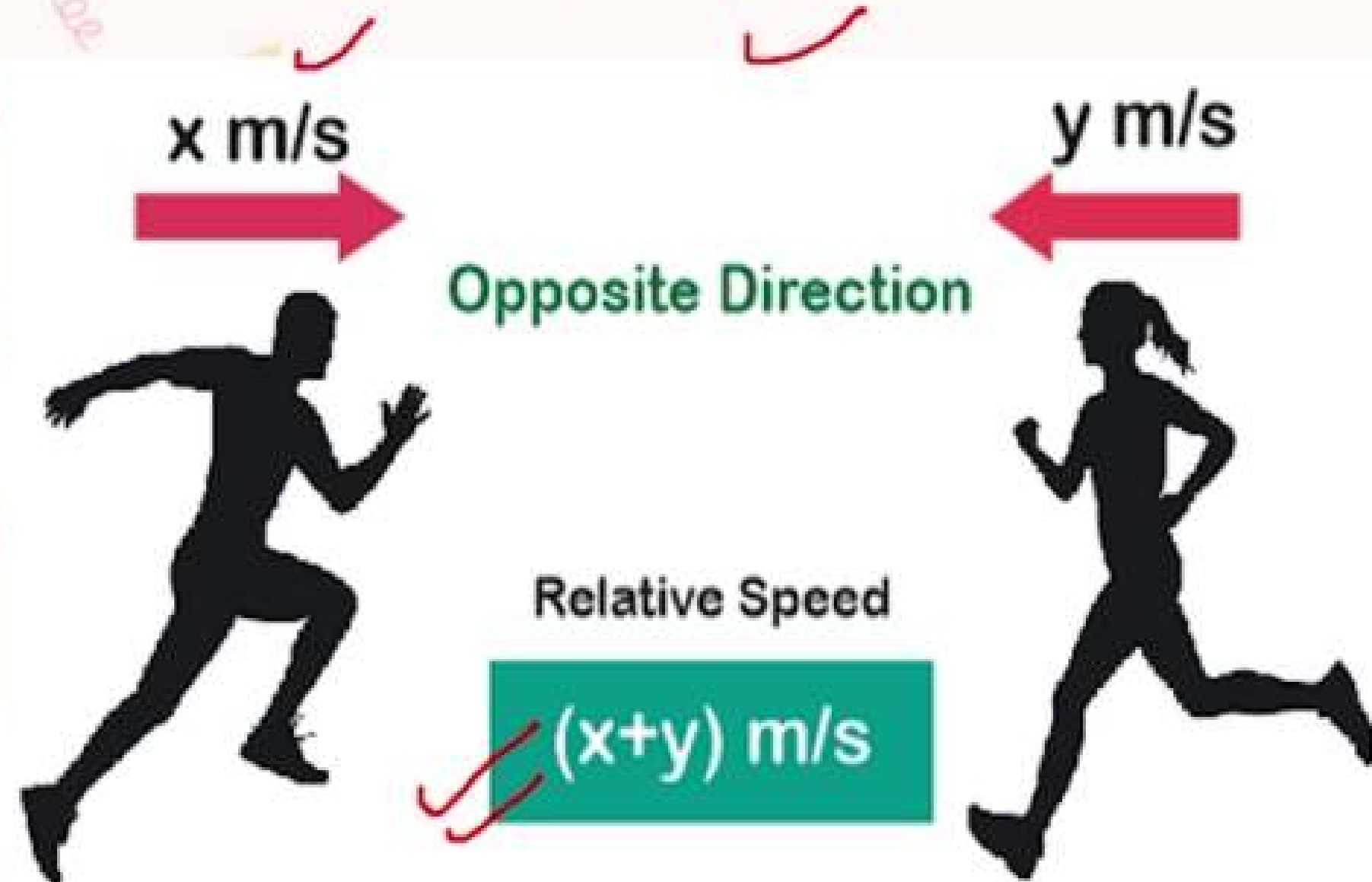


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Relative Speed (After Relationship)



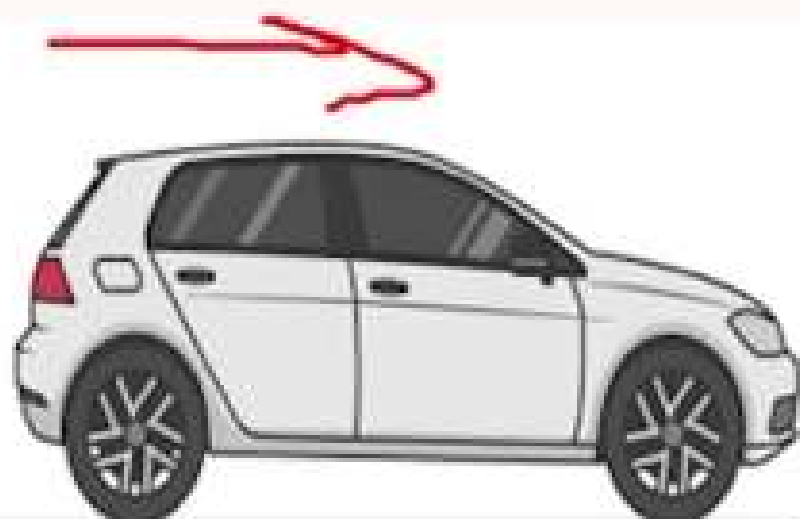
Relative Speed



Relative Speed



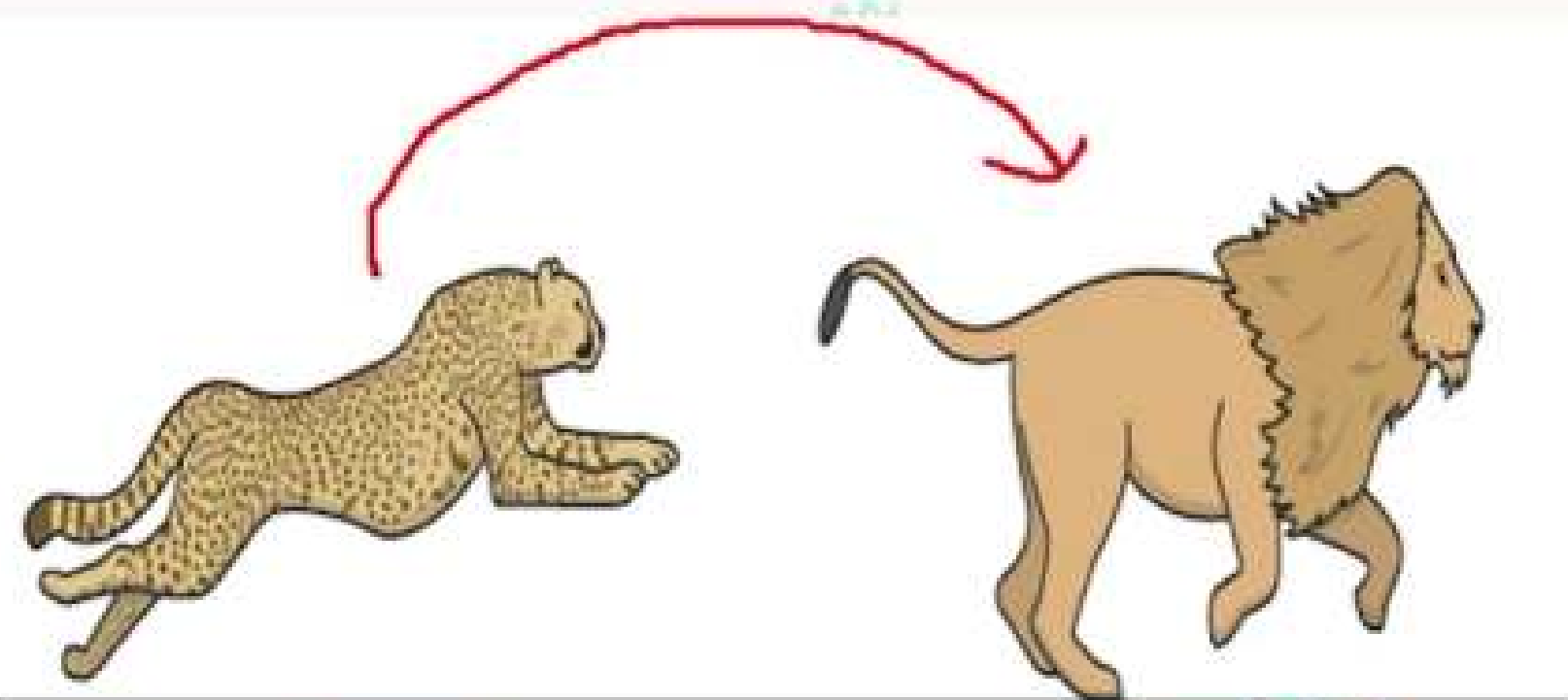
running speed
 $= 3 \text{ m/s}$



driving speed
 $= 25 \text{ m/s}$

relative speed $= 22 \text{ m/s}$

$$25 - 3 = 22 \text{ m/s}$$



42 m/s

40 m/s

$$42 - 40 = \underline{\underline{2 \text{ m/s}}}$$

Relative Speed (After Relationship)

$$1 + 1 = 2$$

$$1 + 1 = 11$$

$$1 + 1 = 1 \leftarrow$$



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Que: Distance between Jetha and Babita Ji is 550 km. Both start walking towards each other with the speed of 60 km/hr and 50 km/hr respectively. Find the time taken by them to meet each other.



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Que: Distance between Jetha and Babita Ji is 550 km. Both start walking towards each other with the speed of 60 km/hr and 50 km/hr respectively. Find the time taken by them to meet each other.



$$S = 60 + 50 \\ = 110$$

$$S = \frac{d}{t}$$

$$t = \frac{d}{S} = \frac{550}{110} \\ = 5 \text{ hours}$$

Que: A boy from Delhi and a girl from Meerut started walking towards each other. They both started with the speed of 5 km/hr. After every 1 hour the boy increased his speed by 1 km/hr and the girl decreased her speed by 1 km/hr. Distance between Delhi and Meerut is 110 km. find the time taken by them to meet each other.



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Que: A boy from Delhi and a girl from Meerut started walking towards each other. They both started with the speed of 5 km/hr. After every 1 hour the boy increased his speed by 1 km/hr and the girl decreased her speed by 1 km/hr. Distance between Delhi and Meerut is 110 km. find the time taken by them to meet each other.

Boy	Girl	Total
5	5	10
6	4	10
7	3	10
8	2	10
9	1	10
10	0	10

✓✓

6 hours → 60 km

Boy	Girl	Total
11	0	11
12	0	12
13	0	13
14	0	14

50 km

4 hours

(6 + 4) = 10 hours